Spring Security is configured to protect all endpoints and resources, so you'll automatically be redirected to the Keycloak login page. After you authenticate as either Isabelle or Bjorn, you'll be redirected back to the Edge Service root endpoint from which the Angular frontend is served.

For now, there's not much you can do. The authentication flow is triggered by Spring Security when it receives an unauthenticated request, but it won't work if it's an AJAX request due to CORS issues. Furthermore, `POST` requests (including the logout operation) will fail due to the CSRF protection enabled by Spring Security. In the following sections, I'll show you how to update the Spring Security configuration to overcome those issues.

Before moving on, stop the application with Ctrl-C (but keep the containers running—you'll need them).

11.5.2 Controlling the authentication flow

In the previous section, you tried accessing the Edge Service homepage and experienced being automatically redirected to Keycloak to provide a username and password. When the frontend consists of server-rendered pages (such as when using Thymeleaf), that behavior works fine, and it's convenient since it doesn't require any extra configuration. If you're not authenticated yet, or your session has expired, Spring Security will automatically trigger the authentication flow and redirect your browser to Keycloak.

With a single-page application, things work a bit differently. The Angular application is returned by the backend when accessing the root endpoint through a standard HTTP `GET` request performed by the browser. After that first step, the SPA interacts with the backend through AJAX requests. When the SPA sends an unauthenticated AJAX request to a protected endpoint, you